

DIRECTIONS *for questions 1 to 4:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Combinatorics is the area of mathematics that is concerned with, relates to, employs, studies combinatorial arguments. So, what is a combinatorial argument?

Explored, known mathematical results can be pictured as a huge imaginary edifice of truths all derivable from one another and from certain original principles. Mathematicians typically try to extend this structure, or apply parts of it to the real world, or both. Such efforts consist of arguments, every argument containing two components: knowledge about the existing structure of mathematics, and *ingenuity*, including complex, devious, and perhaps very detailed and complicated reasoning.

A *combinatorial argument* is one that consists predominantly of ingenuity in detailed reasoning rather than knowledge of existing mathematics. This is in contrast to a *knowledge-based* argument, which relied heavily on piecing together known results. An argument of the following form is not combinatorial: by the theorem of X we have A; by the theorem of Y we have B; from A and B we get the desired conclusion. This kind of argument is based heavily on knowledge of the theorems of X and of Y. It could, however, be a combinatorial argument to you if you were ignorant of the theorems and recreated them yourself.

It is obvious why combinatorial arguments, and by extension, combinatorial subjects were never popular among traditional mathematicians. They spent much of their lives learning lore about their subject. Combinatorial arguments tend to ignore this knowledge. Combinatorics can be attempted by inspired amateurs and dilettantes who know little of formal mathematics, and much of it was developed by such people. One goal of formal mathematics is to systematize the structure of mathematical knowledge to abolish the need for intricate reasoning and thought. Combinatorics therefore represents the areas of mathematics that are failures in this sense, those for which systematic developments of past knowledge are relatively useless in attacking present problems. At one time combinatorics was derisively referred to as the “slums” of mathematics.

Of course, talented mathematicians in all fields routinely use and invent combinatorial arguments. They are honorary combinatorists, differing from true combinatorists mainly in that the problems they work on are far more heavily imbedded in the structure of mathematics than are most combinatorial problems...

When research is performed successfully in any area, results are obtained and knowledge is developed. Eventually all the relatively easy and clearly important results have been discovered, and what is left is difficult or obscure or special or, on first glance, uninteresting. The more we discover, the less is left to find. This development of knowledge is inevitable in all fields, and tends, eventually, to make subjects that have been long and heavily studied forbidding. As a subject matures in this way it becomes necessary for a newcomer in that field to learn more and more existing results to develop the ability to attack problems of less and less interest. In a combinatorial subject, the situation is worse: as it develops, knowledge of past results inevitably grows in importance, eventually becoming as important as ingenuity, and the subject ceases to be combinatorics at all, receding into just one more area of mathematics.

Q1. According to the passage, which of the following is true regarding combinatorics?

- ☐ a) Mathematicians who are associated with traditional mathematics do not use combinatorics in their research.
- ☐ b) The subject of Combinatorics utilises knowledge which is contradictory to the conventional knowledge in mathematics.
- ☐ c) Combinatorics does not use any of the existing knowledge about the structure of mathematics.
- ☐ d) Combinatorics is developed mostly by people who are not professional mathematicians.

Q2. Which of the following is true regarding combinatorial arguments and knowledge based arguments?

- ☐ a) The problems which can be solved using combinatorial arguments are different from the ones that can be solved using knowledge based arguments.
- ☐ b) All the problems that can be solved using combinatorial arguments can also be solved using knowledge based arguments.
- ☐ c) Combinatorial arguments can be used for solving those problems which cannot be solved using knowledge based arguments.
- ☐ d) Knowledge based arguments are easier for the layman to understand as compared to combinatorial arguments.

Q3. Which of the following is the primary reason that combinatorial arguments are not popular among traditional mathematicians?

- ☐ a) Combinatorics highlights the failure of traditional mathematicians in solving the problems which cannot be solved using systematic developments of past knowledge.
- ☐ b) Combinatorics does not make use of most of the systemic knowledge of mathematics which the traditional mathematicians spend most of their lives learning.
- ☐ c) Combinatorics makes use of intricate reasoning and thought with which traditional mathematicians are not comfortable.
- ☐ d) Traditional mathematicians are not comfortable using the subject that amateurs and dilettantes developed.

Q4. If a problem can be solved using the results of three different mathematical theorems, which of the following can be inferred about the problem?

- ☐ a) The problem may be solved using knowledge-based argument but cannot be solved using combinatorial argument.
- ☐ b) The problem may be solved using combinatorial argument but cannot be solved using knowledge-based argument.
- ☐ c) The same problem may be solved by using either knowledge-based argument or combinatorial argument.
- ☐ d) The problem can be solved only by honorary combinatorists but cannot be solved by true combinatorists.

DIRECTIONS *for questions 5 to 8:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Reducing uncertainties in the historical record of Earth's surface temperature can improve scientists' ability to understand and explain changes in the climate over the past 150 years. This is particularly important for the early part of the twentieth century, because the cause of observed warming at that time remains fiercely debated. Writing in *Nature*, Chan et al. demonstrate an innovative approach to account for differences in how sea surface temperature was measured in the early twentieth century...

... The record of global surface temperature is produced by combining measurements of sea surface temperature (SST) with measurements of air temperature over land and ice. The largest remaining uncertainties in the global temperature record are associated with the SST estimates. Specifically, changes in observational instrumentation and techniques over time, coupled with patchy metadata (information about data) and sparse sampling in some regions complicate the interpretation of the historical record.

Initially, SST estimates were made using wooden buckets that were thrown over the sides of ships, filled with water and hauled up. The temperature of the water in the buckets was then measured using a thermometer. While the buckets were being hoisted up, evaporative cooling and exposure to ambient conditions would often reduce the temperature of the water by a few tenths of a degree Celsius.

This bias was exacerbated by a transition to poorly insulated canvas buckets in the late nineteenth century... Accounting for the cold bias in bucket measurements is the single largest adjustment to the ocean (and global) temperature record. Without the adjustment, the estimated rate of ocean warming from 1850 to the present would be about 30% higher.

A bucket measurement can be affected by a wide range of factors. These include the height of the ship, the composition and size of the bucket, how long it remains in the sea, whether the water is stirred before measurement and how long the thermometer is left in the water...

Chan and colleagues found a clever way to tackle this problem. They looked at the difference between SST measurements that were made within 300 kilometres and 2 days of one another, producing a data set of 6 million measurement pairs between 1908 and 1941. Ships were grouped by national origin, on the assumption that ships from the same country would tend to have similar measurement practices at any given time. The authors found sizeable offsets in SST estimates between ship groups, ranging from -0.3°C to $+0.6^{\circ}\text{C}$.

... Chan et al. [also] realized that measurements from Japanese ships in the North Pacific suddenly became about 0.35°C cooler after 1930 when compared with measurements from other countries. This change was caused by the Japanese switching from recording temperatures in whole-degrees Fahrenheit to taking readings in degrees Celsius and then dropping any numbers after the decimal point...

Chan and colleagues' results suggest that scientists have been overestimating warming in the North Atlantic and substantially underestimating warming in the North Pacific during the early twentieth century because of not fully accounting for biases in bucket measurements. These findings bring the difference in estimated warming between the two regions in line with projections from climate models, indicate that intrinsic climate variability has a smaller impact on regional warming rates than was thought...

Q5. Which of the following is an assumption made in the second para of the passage?

- ☐ a) The quality of instrumentation and techniques has improved over time.
- ☐ b) Taking accurate samples is a tedious process.
- ☐ c) Measurements of air temperatures over land and ice have reasonably been more accurate than those of sea surface temperatures.
- ☐ d) We do not have information about how data has been collected in the past.

Q6. The example of measurements from Japanese ships in the North Pacific in the penultimate para of the passage helps establish that

- ☐ a) recording temperatures in whole-degrees Fahrenheit is not as accurate as taking readings in degrees Celsius.
- ☐ b) there wasn't uniformity among nations when it came to ocean temperature measurements.
- ☐ c) approximation is one of the reasons behind inaccurate measurements of SSTs.
- ☐ d) sizeable offsets in SST estimates could exist between groups of ships from one nation.

Q7. It can be inferred from the last para of the passage that

- ☐ a) North Atlantic and North Pacific aren't in the same hemispheres of the earth.
- ☐ b) waters in the North Pacific were warmer in the early twentieth century than what the records suggest.
- ☐ c) variability in the climate does not have any influence on ocean warming rates.
- ☐ d) waters in the North Atlantic were warmer in the early twentieth century than what the records suggest.

Q8. The study published by Chan et al. strengthens the argument that

- ☐ a) ocean warming calculations were prone to inaccuracies because of the methodologies adopted.
- ☐ b) climate models are reasonably inaccurate in their ocean warming projections.
- ☐ c) techniques for measuring ocean temperatures must be standardised.
- ☐ d) the effect of global warming on ocean temperatures is real.

DIRECTIONS for questions 9 to 12: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

... Today's trade tensions are compounding a shift that has been underway since the financial crisis of 2008-09. Cross-border investment, trade, bank loans and supply chains have all been shrinking or stagnating relative to world growth. Globalisation has given way to a new era of sluggishness; we call it "slowbalisation".

During the golden age of globalisation, in 1990-2010, commerce soared as the cost of shifting goods in ships and planes fell, phone calls got cheaper, tariffs were cut, and finance liberalised... As firms set up around the world, investors roamed, and consumers shopped in supermarkets with enough choice ...

Globalisation has slowed from light speed to a snail's pace in the past decade for several reasons. The cost of moving goods has stopped falling. Multinational firms have found that global sprawl burns money and that local rivals often eat them alive. Activity is shifting towards services, which are harder to sell across borders. ...

...Evidence for this is everywhere. Geopolitical rivalry is gripping the tech industry, which accounts for about 20% of world stock markets. ... Tax systems are being bent to patriotic ends...Trade is suffering as firms use up the inventories that they had built up in anticipation of higher tariffs... But what really matters is firms' long-term investment plans, as they begin to lower their exposure to countries and industries that carry high geopolitical risk or face unstable rules. There are signs that an adjustment is beginning. Chinese investment into Europe and America fell by 73% this year. The global value of cross-border investment by multinational companies [also] sank by about 20%.

The new world will work differently. Slowbalisation will lead to deeper links within regional blocs. Supply chains in North America, Europe and Asia are sourcing more from closer to home. In Asia and Europe most trade is already intra-regional, and the share has risen since 2011. Asian firms made more foreign sales within Asia than in America this year. As global rules decay, a fluid patchwork of regional deals and spheres of influence is asserting control over trade and investment...

Yet, slowbalisation has two big disadvantages... Between 1990 and 2010 most emerging countries were able to close some of the gap with developed ones. Now more will struggle to trade their way to riches... Most countries' interest rates will still be affected by America's even as their trade patterns become less linked to it, leading to financial turbulence...

Second, slowbalisation will not fix the problems that globalisation created. Automation means that there will be no renaissance of blue-collar jobs in the West. Firms will hire unskilled workers in the cheapest places in each region. Climate change, migration and tax-dodging will be even harder to solve without global co-operation...

Globalisation made the world a better place for almost everyone. But too little was done to mitigate its costs. The integrated world's neglected problems have now grown in the eyes of the public to the point where the benefits of the global order are easily forgotten. Yet, the solution on offer is not really a fix at all. Slowbalisation will be meaner and less stable than its predecessor. In the end it will only feed the discontent.

Q9. The author's main argument is refuted by which of the following statements?

- ☐ a) Slowbalisation doesn't come at a cost like globalisation did.
- ☐ b) The public has made a conscious decision to pick slowbalisation over globalisation.
- ☐ c) Slowbalisation is not a solution to the problems created by globalisation but a consequence of those problems.
- ☐ d) Slowbalisation in the long run will give way to flourishing of local trade which will help emerging countries grow.

Q10. Which of the following is an adverse consequence of slowbalisation?

- ☐ a) The gap between emerging countries and developed ones will widen.
- ☐ b) Loss of blue-collar jobs because of automation.
- ☒ c) **Financial turmoil caused by the effect of America's interest rates on those of other countries.**
- ☐ d) The rise of impediments in solving migration and climate change issues.

Q11. The author mentions “firms’ long-term investment plans” to demonstrate that:

- ☐ a) there is a deterioration of the global value of cross-border investment by multinational companies.
- ☐ b) the risks of globalisation are increasing for multinational companies.
- ☒ **c) firms are lowering their exposure to countries that pose greater unpredictability. ✓ Your**
- ☐ d) multinational firms are losing out to competition from local companies.

Q12. All of the following are correlated to a new era of sluggishness replacing globalisation EXCEPT:

- ☐ a) There is an increase in intra-regional trade.
- ☐ b) Most countries' interest rates are affected by those of America.
- ☐ c) **Supply chains source more from closer to home.**
- ☐ d) The fall in the costs of shifting goods in planes and ships has been arrested.

DIRECTIONS for questions 13 to 16: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Dine on a steady diet of books these days and it's easy to think that the future belongs to Asia.

Sustained, rapid economic growth since World War II has undeniably boosted the region's economic output and military capabilities. But it's a gross exaggeration to say that Asia will emerge as the world's predominant power player.

Asia is nowhere near closing its economic and military gap with the West. The region produces roughly 30 percent of global economic output, but because of its huge population, its per capita GDP is only \$5,800, compared with \$48,000 in the United States. Asian countries are furiously upgrading their militaries, but their combined military spending in 2008 was still only a third that of the United States. Even at current torrid rates of growth, it will take the average Asian 77 years to reach the income of the average American. And Asia's combined military budget won't equal that of the United States for 72 years.

It is meaningless to talk about Asia as a single entity of power, now or in the future. The fast ascent of one regional player will be greeted with alarm by its closest neighbours. China and Japan have fought repeatedly over Korea; the Soviet Union teamed up with India and Vietnam to check China, while China supported Pakistan to counterbalance India. ... If Asia is becoming the world's center of geopolitical gravity, it's a murky middle indeed.

Those who think Asia's gains in hard power will inevitably lead to its geopolitical dominance might also want to look at another crucial ingredient of clout: ideas. Pax Americana was made possible not only by the overwhelming economic and military might of the United States but also by a set of visionary ideas: free trade, Wilsonian liberalism, and multilateral institutions. Although Asia today may have the world's most dynamic economies, it does not seem to play an equally inspiring role as a thought leader, when it actually should, given the circumstances. The big idea animating Asians now is empowerment; Asians rightly feel proud that they are making a new industrial revolution. But self-confidence is not an ideology, and the much-touted Asian model of development does not seem to be an exportable product.

Given Asia's relatively low per capita income, its growth rate will indeed outpace the West's for the foreseeable future. But the region faces enormous demographic hurdles in the decades ahead. More than 20 percent of Asians will be elderly by 2050. ... China's elderly population will soar in the middle of the next decade. Its savings rate will fall while healthcare and pension costs explode. ...

Environmental and natural resource constraints could also prove crippling. Pollution is worsening Asia's shortage of fresh water while air pollution exacts a terrible toll on health. Without revolutionary advances in alternative energy, Asia could face a severe energy crunch. Climate change could devastate the region's agriculture.

The current economic crisis, moreover, will lead to huge overcapacity as Western demand evaporates. Asian companies, facing anemic consumer demand at home, will not be able to sell their products in the region. The Asian export-dependent model of development will either disappear or cease to be a viable engine of growth.

Q13. The tone of the author towards the rise of Asia can be termed as one of

- ☐ a) great fascination.
- ☐ b) strong scepticism.
- ☒ c) **pointed criticism.** ✖ Your answer is incorrect
- ☐ d) guarded appreciation.

Q14. It can be inferred from the passage that the prime reason(s) for the rise of the United States as a global superpower was/were

a. its vision that manifested itself in free trade, Wilsonian liberalism and multilateral institutions.

b. its huge population responsible for intellectual wealth and its dynamic economy capable of rising to the occasion when needed.

c. its military mastery and economic domination in the post World War II era.

d. America being the first to harness nuclear power.

☐ a) (b) and (d)

☐ b) (c) and (d)

☐ c) **(a) and (b)**

☐ d) (a) and (c)

Q15. The author mentions some stumbling blocks in the course of the expected dawn of a new Asian age. Three of the following choices mention these blocks. Pick the exception.

- ☐ a) Asia's lack of visionary ideas and its export-dependent model of development.
- ☐ b) Competition for power among neighbours in the region.
- ☐ c) **Asia's relatively low per capita income.**
- ☐ d) Asia's environmental concerns and demographic hindrances.

Q16. Which of the following suggestions associated with 'thought leadership' would the author most likely advocate for Asia to enhance its superpower status?

- ☐ a) It should continue to use the ideology of empowerment.
- ☐ b) It should employ broad, visionary ideas that will shift the geopolitical lever in its favour.
- ☐ c) It should use its intellectual capability in promoting a 'cohesive identity'.
- ☒ d) It should use the concept of thought leadership in converting its export-dependent model of development into other suitable models. **X You**

Q17. DIRECTIONS *for questions 17 and 18:* The passage given below is followed by four summaries. Choose the option that best captures the essence of the passage.

Those who believe in 'art for art's sake' might say that art as an unquantifiable force must remain outside social or ideological norms, or risk becoming something else. Some experts argue that it's a slippery slope when art starts leaning toward activism because that's just not the point. It's a viewpoint committed to art as a force on its own, a process of radical experimentation that results in an artwork, one of many along a line of inquiry, not a means to illustrate an end or impregnate an object with meaning. No conclusions should be drawn about art, present or future because it is the force against universalism, which must be interrupted by artists, as if to tell the world "wake up!"

- ☐ a) Art, in itself, doesn't have a universal meaning and, hence, it should not be used for propaganda of any sort.
- ☐ b) The purpose of art is to extend the process of reflection, and not to represent an end goal.
- ☐ c) **Art is constantly evolving, and no artwork can have a comprehensive meaning.**
- ☐ d) **Art cannot be used for activism because one cannot draw conclusions about art.**

Q18. DIRECTIONS *for questions 17 and 18:* The passage given below is followed by four summaries. Choose the option that best captures the essence of the passage.

For Pacific islanders, both land and ocean have a huge economic, political and cultural significance. The region has an ocean-based economy focused on maritime transportation and shipping, fisheries, extractive industries and prospecting for oil and gas as well as tourism. In 2020 alone, the revenue generated through fishing licensing stood at US\$474 million, which ever since has been growing steadily. Tourism industry, where the tourists flock to these unspoilt natural beauties of the islands, generates up to 70 per cent of any Pacific country's gross domestic product (GDP). Agriculture is the other vitally important sector that generates wealth and employment to the rural population. Climate change and severe weather events severely impact all these sectors and leave behind destitute populations and aid-dependent countries.

- ☐ a) Pacific islanders would be impacted more than anyone because of severe weather events and climate change.
- ☐ b) The ocean dependency of Pacific countries makes the population highly vulnerable to climate change.
- ☐ c) **Pacific countries that depend on tourism, agriculture and the ocean for their wealth are being devastated by climate change.**
- ☒ d) **Owing to their economy's dependence on land and ocean, Pacific islanders are especially susceptible to climate change events.** ✓ **Your answer is correct**

Q19. DIRECTIONS *for questions 19 and 20:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.
Sentence: Some say the Romantics opposed reason; others simply think of candlelit dinners and passionate declarations of love.

Paragraph: Today, the term ‘Romanticism’ evokes images of lonely figures in moonlit forests or on craggy cliffs – as expressed in Caspar David Friedrich’s paintings – as well as artists, poets and musicians who emphasised emotion and longed to be at one with nature. _____(1)_____ For the Jena Set, though, Romanticism was something much more complex and radical. Romantic poetry, they said, was unruly and dynamic – a ‘living organism’. _____(2)_____ They wanted to romanticise the entire world. They strived to unite humankind and nature, art and science. If two elements could create a new chemical compound, so Romantic poetry could bring together different disciplines and subjects, and weld them into something new. _____(3)_____ ‘By giving the commonplace a higher meaning,’ Novalis said, ‘by making the ordinary look mysterious, by granting to what is known the dignity of the unknown and imparting to the finite a shimmer of the infinite, I romanticise.’ _____(4)_____ And for that, the friends insisted, one needs imagination.

- ☐ a) Option 1
- ☐ b) Option 2
- ☐ c) **Option 3**
- ☐ d) **Option 4**

Q20. DIRECTIONS *for questions 19 and 20:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.
Sentence: Of course, the number of daylight hours on a given day is fixed, and setting the clocks ahead one hour does nothing to increase the amount of daylight.

Paragraph: One interesting example of a government-imposed self-control strategy is daylight saving time (or summertime, as it is called in many parts of the world). _____(1)_____ Surveys reveal that most people think that daylight saving time is a great idea, primarily because they enjoy the “extra” hour of daylight during the evening. _____(2)_____ The simple change of the labels on the hours of the day, calling “six o’clock” by the name “seven o’clock,” nudges us all into waking up an hour earlier. _____(3)_____ Along with having more time to enjoy an evening softball game, we end up saving energy too. _____(4)_____ Historical note: the idea was first suggested by Benjamin Franklin during his tenure as an American delegate in Paris. A well-known skinflint, Franklin calculated that thousands of pounds of candle wax could be saved with his idea. However, the idea did not catch on until World War I.

- ☐ a) Option 1
- ☐ b) Option 2
- ☐ c) **Option 3**
- ☐ d) **Option 4**

Q21. DIRECTIONS *for questions 21 and 22:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Most of the mercury that gets trapped in wetlands sits in sediments in its oxidized form, either as mercury sulphide or just bound to organic matter.
2. But photochemical reactions due to sunlight can free mercury floating around in the water of lakes and the surface of the ocean as well as from dry soils, turning oxidized mercury attached to organic matter into elemental mercury.
3. In this form, it stays in the ground and can't escape.
4. In this form, the metal can volatilize.

Q22. DIRECTIONS *for questions 21 and 22:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Captive birds provide a useful model for investigating aspects of social choice in highly gregarious, long-lived species.

2. Captive populations therefore provide a useful alternative for measuring social choices.

3. Animals invest in social relationships that convey fitness benefits and bonds can be long-lasting.

4. For some species, field-based measurement of social networks can be difficult.

Q23. DIRECTIONS *for question 23:* The passage given below is followed by four alternate summaries. Choose the option that best captures the essence of the passage.

Essentially, multipolarity means that instead of speaking a common language, the major poles speak different policy languages. Trade-based tension is an obvious possibility here. Another form of tension is the crisis of identity created for countries that are not wholly within one of the poles—again, Japan, Australia, and the United Kingdom are the prime examples—and the crisis of ambition for countries, such as Russia, that want to be poles but lack the wherewithal to do so convincingly.

- ☐ a) Multipolarity opens up the possibility of tensions, related to trade, or to identity or ambition of countries that aren't part of the major poles.
- ☐ b) Multipolarity creates tensions between countries that do not want to align with the major poles.
- ☐ c) **Multipolarity involves tensions due to disagreement in policy, trade, or in ambition and identity.**
- ☐ d) **Multipolarity is created when major forces disagree on policy leaving out the countries which do not have the clout.**

Q24. DIRECTIONS *for question 24:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. The study of these systems therefore provides valuable insight into nonlinearities and the emergence of causal structures in quantum mechanics—essential for any formulation of a quantum theory of gravity.

2. Closed timelike curves are among the most controversial features of modern physics.

3. However, in the quantum regime these paradoxes can be resolved, leaving closed timelike curves consistent with relativity.

4. As legitimate solutions to Einstein’s field equations, they allow for time travel, which instinctively seems paradoxical.